Compound Interest

Compound interest is different from simple interest in that you earn interest not only on the principal but also on the previous interest. Let's look at the following example:

Simple interest:
You borrow \$2000 at a simple interest rate of 5.5% for 5 years. The simple interest formula means you

Find the interest for one year:	0.055(2000) = \$110.00
You pay this interest for each year:	5 years X \$110 = \$550.00

So you will pay \$550 in interest. The total you will pay back is \$2000 + \$550 = \$2550.00.

This is the type of interest you pay on money you borrow. If you pay the loan off early, say in four years, you don't owe the interest for the 5th year.

Compound interest means you pay (or earn) interest not only on the principal but also on the interest. So, for the same problem we might have the following table if the interest is calculated on a yearly basis:

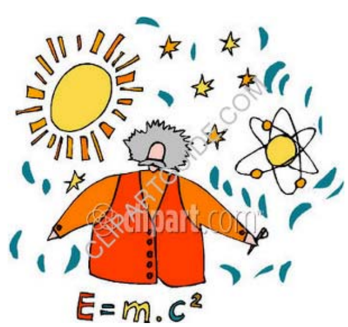
First year:	\$2000 X 5.5% interest = \$110 means you owe 2000 + 110 = \$2110
Second year:	\$2110 X 5.5% interest = \$116.05 means you owe \$2110 + 116.05 = \$2226.05
Third year:	\$2226.05 X 5.5% interest = \$122.43 means you owe \$2226.05 + 122.43 = \$2348.48
Fourth year:	\$2348.48 X 5.5% interest = \$129.17 means you owe \$2348.48 + 129.17 = \$2477.65
Fifth year:	\$2477.65 X 5.5% interest = \$136.27 means you owe \$2477.65 + 136.27 = \$2613.92

This is a difference of \$63.92 over the simple interest plan. You can see that you would NOT want to owe money paying compound interest! And, compound interest makes your savings grow at a faster rate. In college algebra you will learn about continuous interest – continuous interest REALLY makes the money grow!

Einstein said that the most powerful force on earth is compound interest!

Now, lets look at the formula for compound interest: $A = P\left(1+\frac{r}{n}\right)^{nt}$

Each letter has the following meaning:



A = the amount you will have in your account after the specified time.
P = the Principal or the amount of money you start with in the account.
r = the rate of interest
n = the number of compoundings
t = the time in years

Let's look at an example:

Stephanie puts \$1000 in an account to save for her graduation trip. Her bank pays 2.5% compounded quarterly. How much money will she have in 3 years?

P = the money Stephanie starts out with = \$1000
r = the rate of interest = 2.5%
n = the number of compoundings = 4 since there are 4 quarters in one year
t = the time in years = 3

Now substitute into the equation to get A = the amount she will have at the end of 3 years:

$$A = P\left(1 + \frac{r}{n}\right)^{nt} = 1000\left(1 + \frac{0.025}{4}\right)^{4(3)}$$

The rest is simplification using the order of operations:

$$A = P\left(1 + \frac{r}{n}\right)^{nt} = 1000\left(1 + \frac{0.025}{4}\right)^{4(3)} = 1000(1 + 0.00625)^{4(3)}$$

$$A = 1000(1.00625)^{12)} = 1000(1.077632599) = \$1077.63$$

So she will have $1077.63 in the account at the end of 3 years.

The number of compoundings is an important concept in this formula. This is the number of times in a year that the bank will perform the calculations (like in the first example). The formula makes it faster to calculate. So remember these:

Quarterly = 4 times a year
Monthly = 12 times a year
Weekly = 52 times a year
Daily = 360 times a year
Annually = 1 time a year

Suppose Stephanie put the money in a CD that was earning 4.25% compounded quarterly? How much more money would she have?

$$A = P\left(1 + \frac{r}{n}\right)^{nt} = 1000\left(1 + \frac{0.045}{4}\right)^{4(3)} = 1000(1 + 0.01125)^{4(3)}$$

$$A = 1000(1.01125)^{12} = 1000(1.143674441) = \$1143.67$$

Not bad!

It is also important NOT to round off when you do the division and raising to the power. Each time you round off you will cheat yourself of money!

Special Keys to help with Compound Interest Problems

If you are using a scientific calculator, you should find a key that looks like this: y^x or x^y. This is a key that will help you raise numbers to a large power.

For example, say you want to raise 4 to the 6^{th} power (4^6). You would use the following key strokes:

4 $\boxed{y^x}$ 6 = You would then see the display/result of 4096.

So, if you are doing a compound interest formula and you have something like this:

Sarah puts $1000 in an account earning 3% interest compounded quarterly. How much will be in the account after 3 years?

A = what you want to find ($ she will have in 3 years)
P = 1000
R = 3%
T = 3
N = 4

Substitute into the compound interest formula to get:

$$A = 1000\left(1 + \frac{.03}{4}\right)^{12}$$

Using the calculator, enter the following steps:

.03 ÷ 4 = and you will see .0075
Do NOT clear the calculator at this step but now push the following keys:

+ 1 = and you will see 1.0075
Do NOT clear the calculator at this step but now push the following keys:

$\boxed{y^x}$ 12 = and you will see 1.093806898

Do NOT clear the calculator at this step but now push the following keys:

X 1000 = and you will see 1093.806898

Since we are looking for a money answer, round to 2 decimal places to get the answer: $1093.81.

So Sarah will have $1093.81 in her account after 3 years.

Never round between steps! This will really throw the answer off!